

DELHI PUBLIC SCHOOL
SUBJECT- MATH CLASS-8

- Q1.** What does it mean for two or more related quantities to be in a "proportional relationship"?
- Q2.** In the context of geography and map reading, explain what a "Representative Fraction" (RF) indicates.
- Q3.** If you are given a ratio with four terms, such as $a:b:c:d$, what must happen to each quantity to maintain that proportional relationship?
- Q4.** Define "direct proportion" and explain how the quotient of two such quantities behaves.
- Q5.** Briefly describe the concept of "inverse proportion" and how one quantity reacts when the other increases by a specific factor.
- Q6.** Explain the cross-multiplication method used to verify if two ratios, such as $a:b$ and $c:d$, are proportional to one another.
- Q7.** When constructing a pie chart, how is the specific angle for each "slice" determined in relation to the data it represents?
- Q8.** A map has an RF of 1:50,000. If you measure 4 cm between two points on the map, explain the steps you would take to find the actual ground distance in kilometers.
- Q9.** Describe the mathematical process for dividing a whole number, x , into a ratio containing three terms ($a:b:c$).
- Q10.** Using the example of speed and time for a fixed distance, explain why their product remains constant ($xy=k$) in an inverse relationship.
- Q11.** A chef is famous for a sauce mixed in the ratio of Oil: Vinegar: Honey :: 5:2:1. If the chef accidentally uses 150 ml of Honey, calculate the required amounts of Oil and Vinegar needed to keep the flavor profile exact, and find the total volume of the resulting sauce.
- Q12.** A construction crew of 12 people can complete a renovation project in 10 days. If the client demands the work be finished in only 4 days, use the principles of inverse proportion to determine how many additional workers must be hired.
- Q13.** Two pipes are used to fill a swimming pool. Pipe A can fill the pool alone in 4 hours, and Pipe B can fill it alone in 6 hours. Calculate how long it will take to fill the pool if both pipes are opened simultaneously.
- Q14.** You are tasked with drawing a pie chart for a school's annual budget. The spending is categorized as: Academics (40%), Sports (30%), Maintenance (20%), and Administration (10%). Calculate the exact central angle required for each sector and explain the importance of the HCF in simplifying these calculations.
- Q15.** Analyse the relationship between the number of periods in a school day and the duration of each period. If a school currently has 6 periods of 60 minutes each, explain mathematically why increasing the number of periods to 8 (while keeping the total school hours constant) represents an inverse proportion. Calculate the new duration per period.